Year 12 Further Maths - Applications of matrices

Similar questions - Reflections, invariant lines and successive transformations

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The matrix $\mathbf{A} = \frac{1}{5}\begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}$ represents a transformation T which is a reflection in a straight line through the origin O.

(a) Explain why there must be a line of invariant points for T, and state its geometrical significance. **[2]**

(b) By considering the line of invariant points, determine the equation of the mirror line in the form $y = mx + c$. **[4]**

(c) P is the point $(2, 6)$. Determine the coordinates of the point on the mirror line which is closest to P. **[3]**

(d) The line $y = ax + 1$ is an invariant line for T ($a \neq 0.5$). Find a. **[3]**

2 Three transformations T_A, T_B and T_C are represented by $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} a & -1 \\ 1 & a \end{pmatrix}$ (a real).

(a) Find the matrix that represents the inverse transformation of T_A. **[2]**

(b) Determine whether T_A followed by T_B is the same as T_B followed by T_A. **[2]**

(c) R is (T_A followed by T_B) followed by T_C and S is T_A followed by (T_B followed by T_C). Explain why R and S are the same transformation. **[2]**

(d) A quadrilateral Q with vertices D, E, F, G in anticlockwise order has area 5. Under R it becomes Q'. (i) State the order of the vertices of Q' starting from D'. (ii) Find the area of Q' in terms of a. (iii) Explain whether the inverse transformation of R exists. **[5]**

Question 1

(a) A reflection fixes every point on the mirror line, so T has a line of invariant points, and this line is the mirror line

(b) $\begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}\binom{x}{y} = 5\binom{x}{y}$: $3x+4y=5x \Rightarrow y=\frac{1}{2}x$ (the second row gives the same). Mirror line $y=\frac{1}{2}x$ ($m=\frac{1}{2}$, $c=0$)

(c) Closest point is the foot of the perpendicular from P: project $(2,6)$ onto $(2,1)$: $\dfrac{4+6}{5}=2$ so the point is $2(2,1)=(4,2)$

(d) Invariant lines other than the mirror line are perpendicular to it (gradient -2): lines parallel to the mirror line are mapped to a different parallel line. So $a=-2$

Question 2

(a) $\mathbf{A}^{-1}=\dfrac{1}{2}\begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix}$

(b) T_A then T_B has matrix $\mathbf{BA}=\begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix}$, T_B then T_A has matrix $\mathbf{AB}=\begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$. These are different, so not the same

(c) Matrix multiplication is associative: $\mathbf{C}(\mathbf{BA})=(\mathbf{CB})\mathbf{A}$

(d)(i) $\det\mathbf{B}=-1<0$ and $\det\mathbf{A}=2>0$, $\det\mathbf{C}=a^2+1>0$, so $\det(\mathbf{CBA})=-2(a^2+1)<0$: orientation reverses, so the order is clockwise: D', E', F', G' read clockwise. (ii) Area $=5\times 2(a^2+1)=10(a^2+1)$. (iii) $\det=-2(a^2+1)\neq 0$ for all real a, so R has an inverse